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Abhirup Sarker

Lecturer

Department of Robotics and Mechatronics

Engineering

Daffodils International University



MULTIPLEXING

What is multiplexing?

Multiplexing in data communications is a method that combines multiple signals or data streams into one signal over a shared medium. So, Multiplexing allows the simultaneous transmission of multiple signals across a single data link. It basically enable efficient use of bandwidth.



Bandwidth is the range of frequencies a communication channel or signal occupies. It indicates how much frequency space is available to carry information.

Bit rate signifies the number of bits transmitted per second. It reflects how fast data is transmitted.

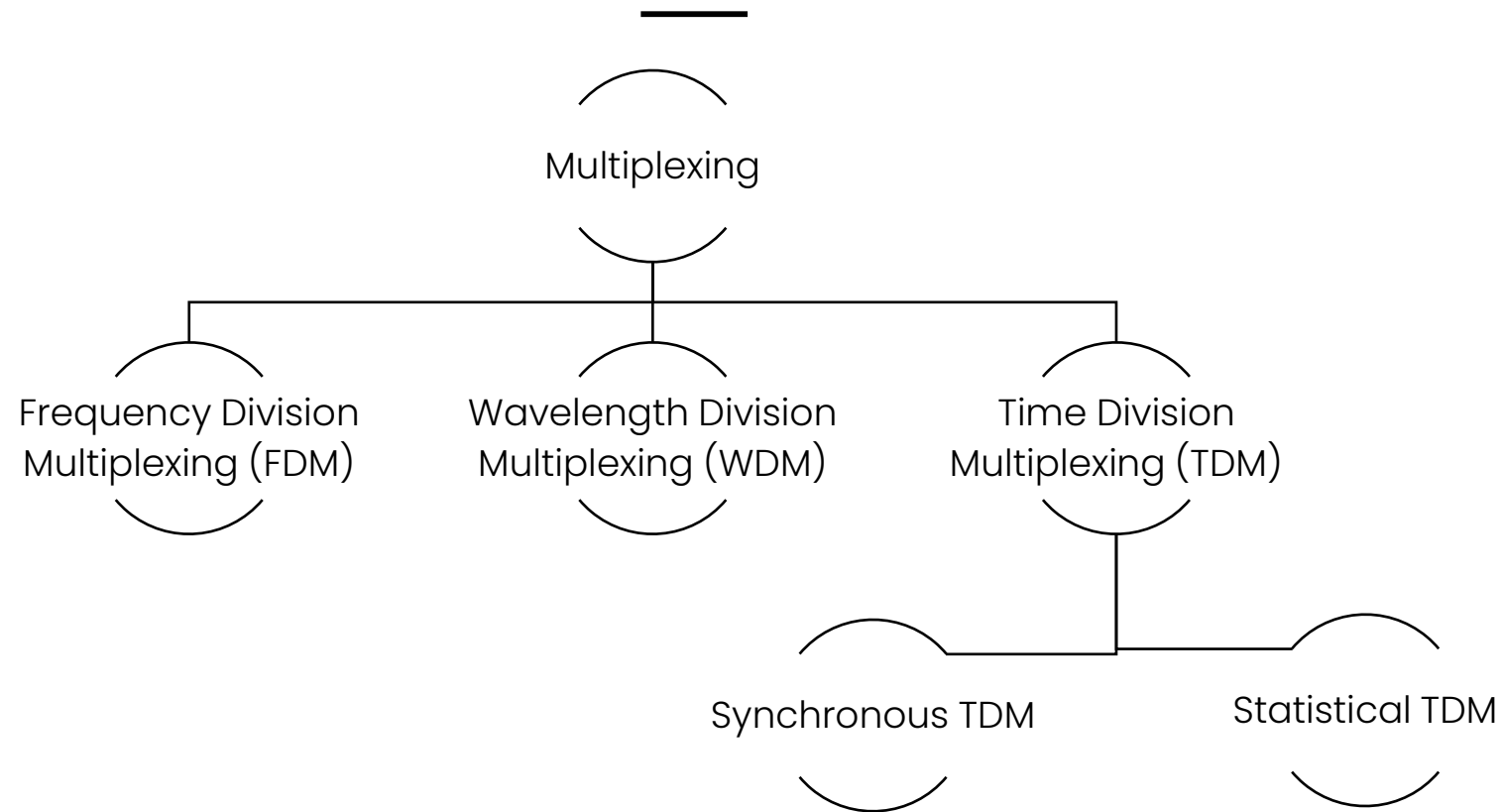
Baud rate is the number of signal changes or symbols sent per second over a communication channel.

DEMULTIPLEXING

Demultiplexing is the process of separating combined signals or data streams back into their original individual channels at the receiver. In simple terms it is the opposite of multiplexing

- It uses bandpass filters to separate the individual signals by selecting one channel
- Then uses demodulator to recover original message
- Multiplexing is many to one process and demultiplexing is one to many process

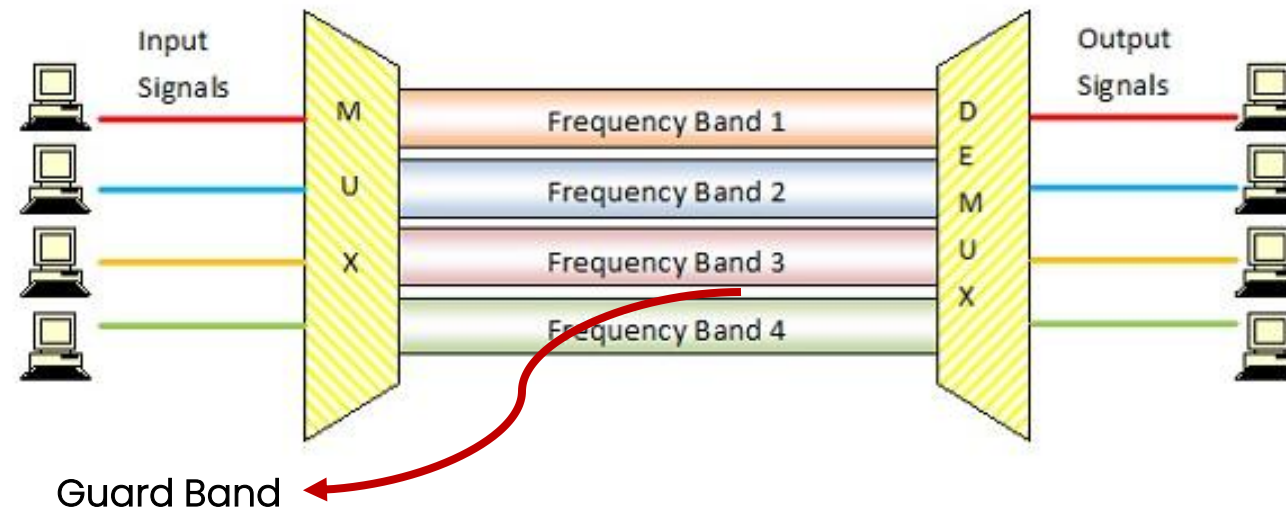
TYPES OF MULTIPLEXING



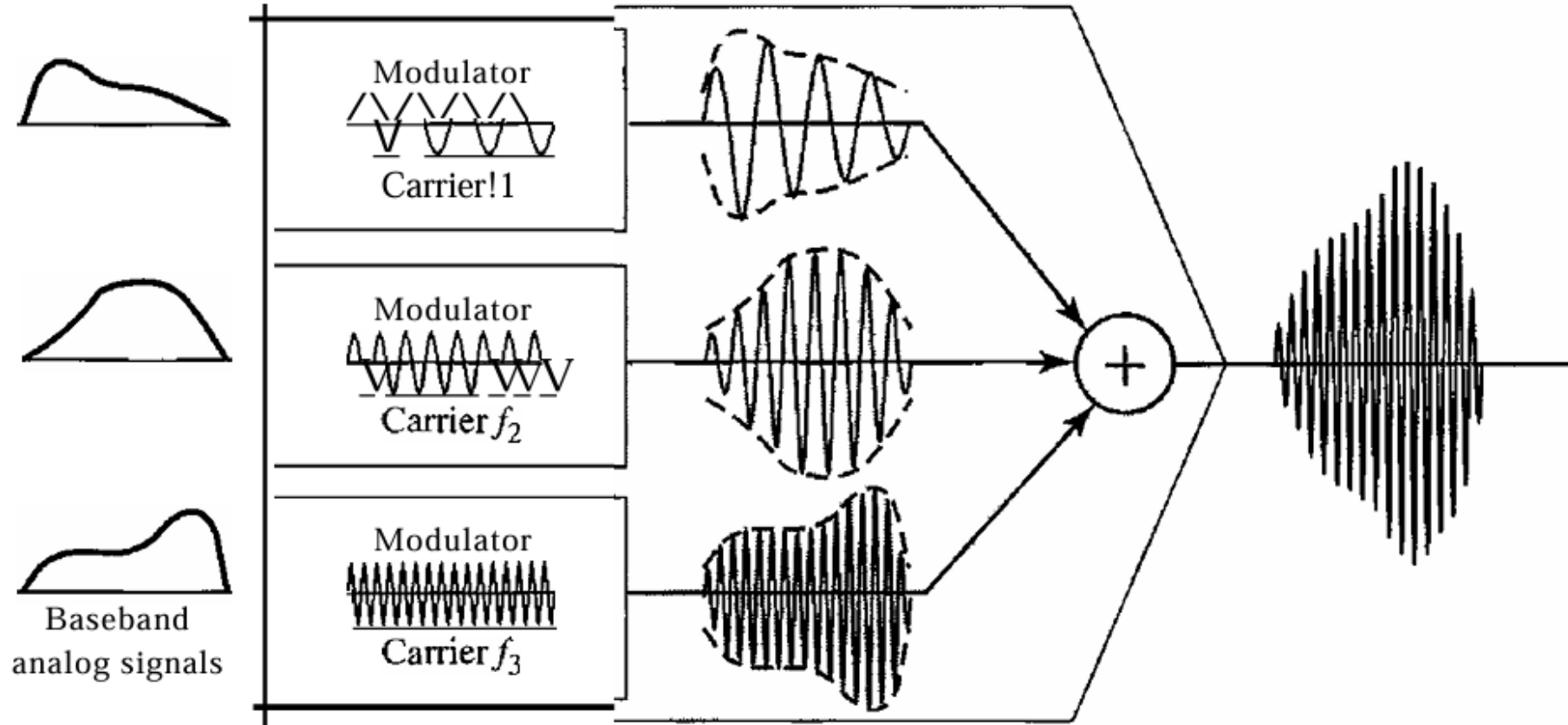
FDM and WDM are analog multiplexing methods, they work with analog signals. Meanwhile, TDM is digital multiplexing technique.

FREQUENCY DIVISION MULTIPLEXING

- Frequency division multiplexing is defined as a type of multiplexing where the bandwidth of a single physical medium is divided into a number of smaller, independent frequency channels.
- FDM is applied when the bandwidth of a link/medium is greater than the combined bandwidths of the signals to be transmitted.
- In FDM, there is chance of inter-channel cross-talk as the bandwidth is divided into multiple frequency channels.
- In order to prevent the inter-channel cross talk and overlapping, unused strips of bandwidth must be placed between each channel. These unused strips between each channel are known as guard bands.

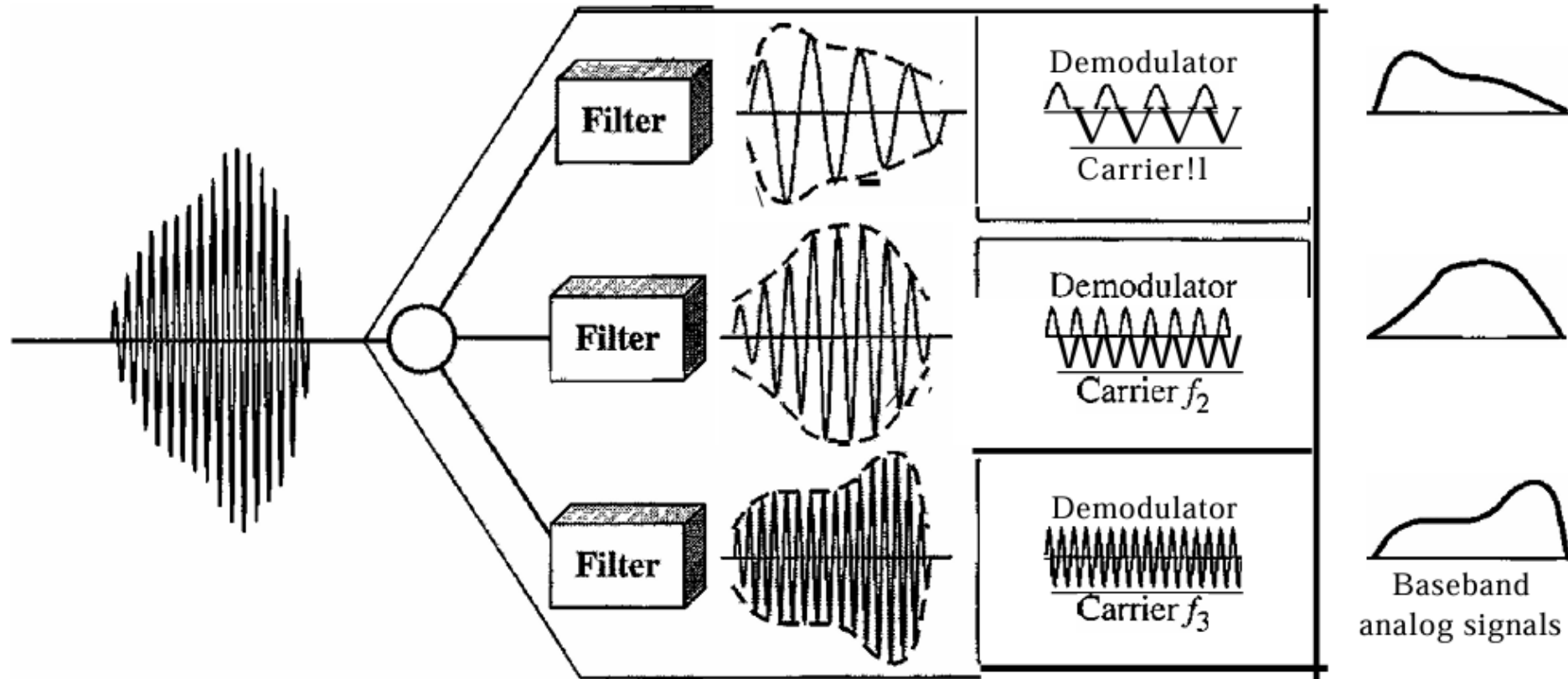


FREQUENCY DIVISION MULTIPLEXING



We are sending multiple modulated signals over a single medium. Multiplexer prepares resultant signal that goes to receiver end.

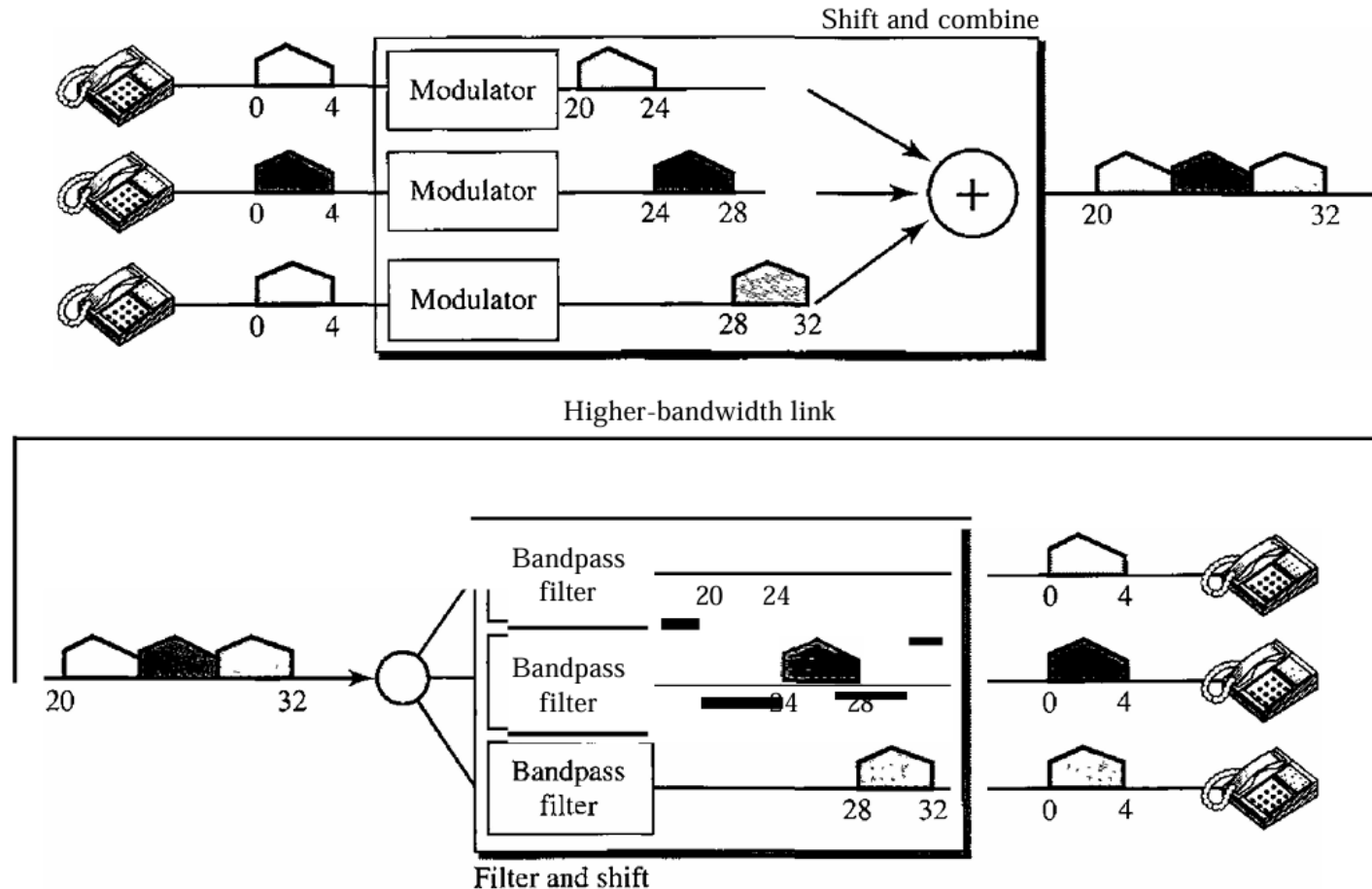
FREQUENCY DIVISION MULTIPLEXING



Demultiplexer filters and separates the signals. The separated signals are demodulated to retrieve original signal/message.

FREQUENCY DIVISION MULTIPLEXING

Problem 1: Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz. Let the channel bandwidth be from 20 to 32 kHz. Assuming there are no guard bands, show and explain the configuration.



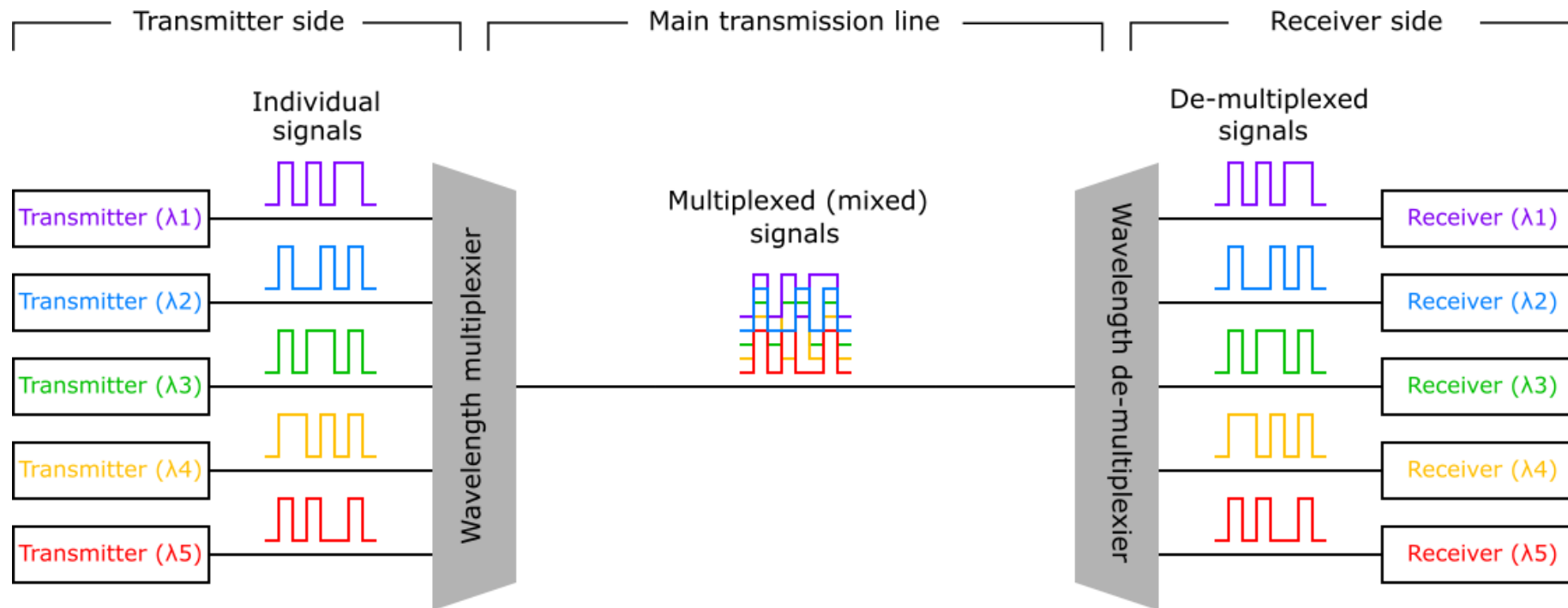
FREQUENCY DIVISION MULTIPLEXING

Problem 2: Assume five channels, each with a 100-kHz bandwidth, are to be multiplexed together. There is also need for guard band of 10 kHz between the channels to prevent overlapping. What is the minimum bandwidth of the link/medium?

Solve Yourself

WAVELENGTH DIVISION MULTIPLEXING

Wavelength Division Multiplexing is a multiplexing technology used to increase the capacity of optical fiber by transmitting multiple optical signals of different wavelengths simultaneously over a single optical fiber.



WAVELENGTH DIVISION MULTIPLEXING

There are two types of WDM:

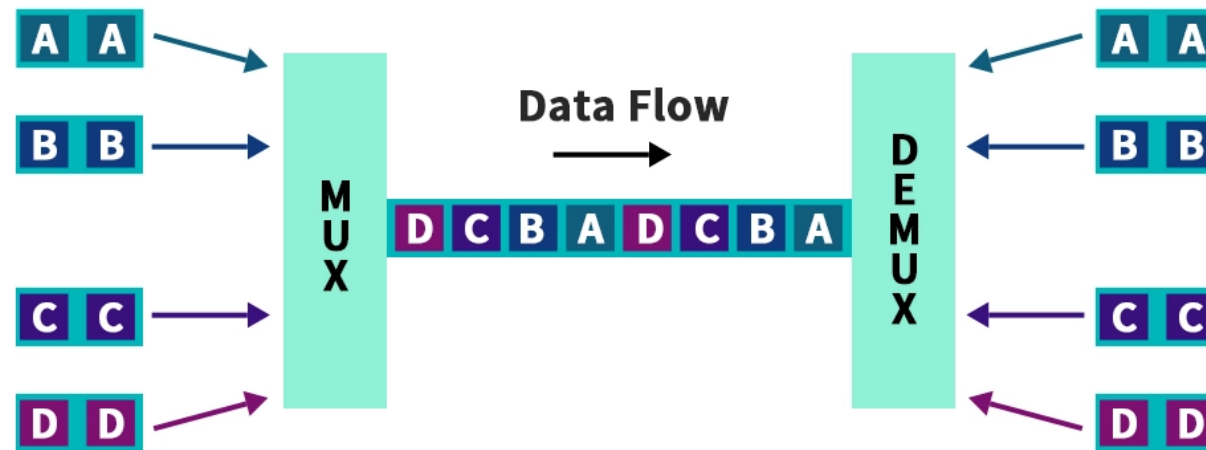
- I. **Dense WDM:** It is used to multiplex a large number of optical signals onto a single optical fiber. DWDM is used to carry different wavelengths that are very close to each other. It overcomes the speed and distance limitations of optical fiber and prevent attenuation as well.
- II. **Coarse WDM:** It is used for lower-capacity applications compared to DWDM. The working principle is similar as it also combines different wavelengths to transmit over optical fiber but can not send large amount of information simaltenously.

FDM VS WDM

Frequency Division Multiplexing	Wavelength Division Multiplexing
It divides bandwidth into smaller frequency signals	It combines several light beams of different wavelength
FDM works with electrical signal or radio frequency	WDM works with optical signal
Transmission medium is copper wire, radio waves	WDM transmission medium is optical fiber
Application is radio, television, telephone	Application in internet

TIME DIVISION MULTIPLEXING

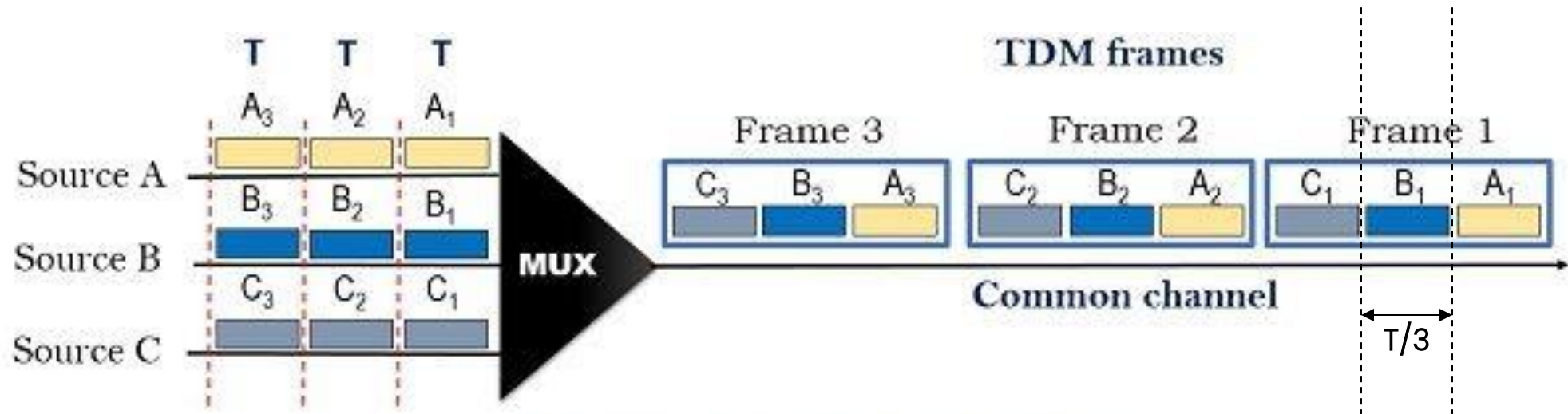
In TDM each sender is given the entire possession of the whole bandwidth of the channel for a fixed duration of time. It combines several low rate channels into one high rate channel. Mainly two types: i. Synchronous TDM and ii. Statistical TDM.



Time Division Multiplexing

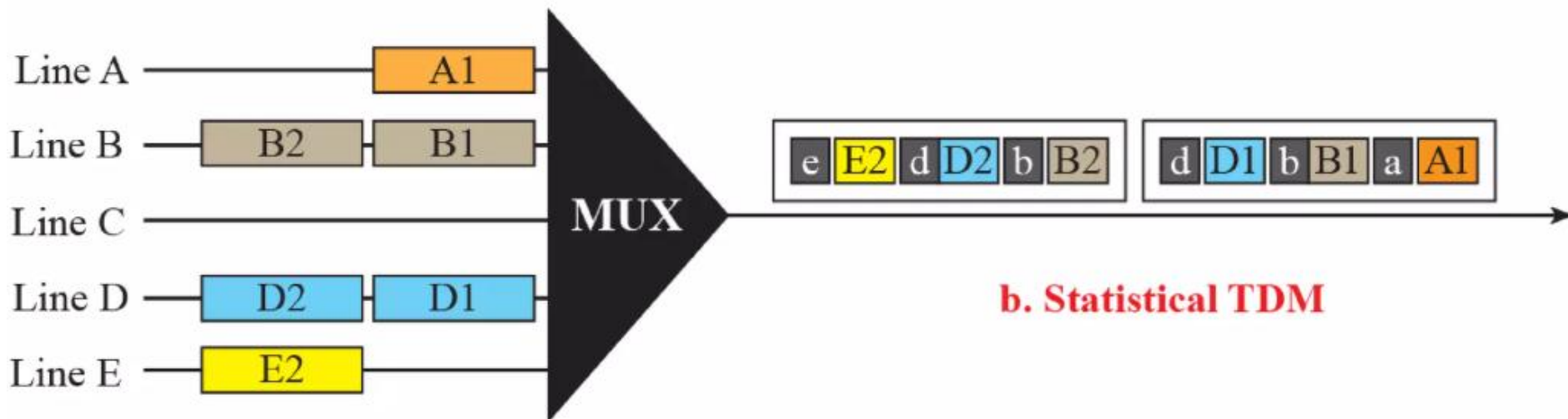
SYNCHRONOUS TDM

- ❑ In synchronous time division multiplexing data is divided into units called frame and each input have time slot to transmit data.
- ❑ If there are n number of participants then required slot in a frame will also be n.
- ❑ If data is taken at T interval, then the slot duration will be T/n .
- ❑ The slots remain empty if the source is not ready to send data, which is inefficient.



STATISTICAL TDM

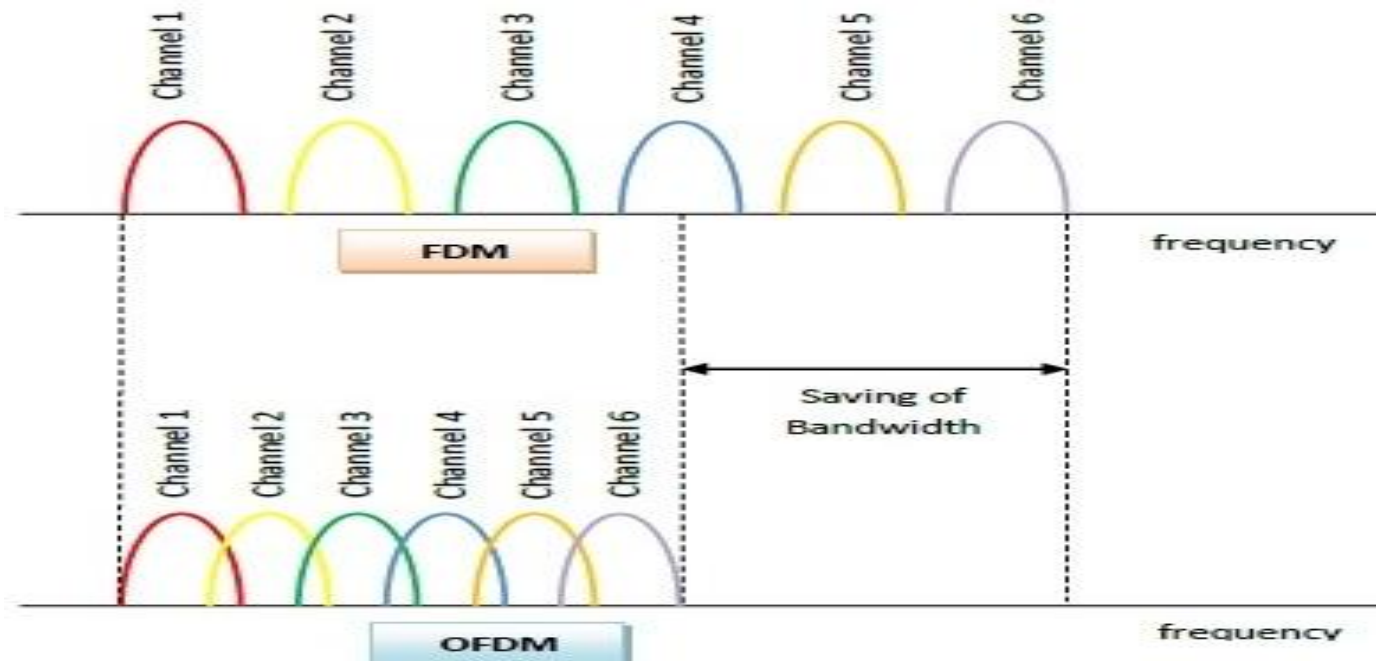
- ❑ In statistical TDM slots are dynamically allocated to improve the bandwidth efficiency.
- ❑ When source is ready to transmit data, it is given a slot in carrying frame.
- ❑ As the data is not synchronous, we attach a header or destination address in front of the data.
- ❑ This method is more efficient as resources are not wasted.



b. Statistical TDM

ORTHOGONAL FREQUENCY DIVISION MULTIPLEXING

- OFDM is a digital multiplexing and modulation technique where data is transmitted simultaneously over many closely spaced orthogonal subcarriers.
- FDM is used for analog signals but with advancing technology and introduction to new digital systems OFDM has gained popularity.
- In OFDM the signals overlap each other unlike FDM where, band guard is necessary to avoid crosstalk
- The orthogonal nature of the signal eliminates inter-channel cross-talk.
- It uses the bandwidth more efficiently and provides high data rate.



REFERENCE

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Data Communications and Networking by Behrouz A. Forouzan
Example: 6.1, 6.2, 6.6, 6.7, 6.8, 6.9



THANK YOU

